

Leading Kazhdan–Lusztig coefficients with fully commutative lower endpoints in type E^*

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Abstract

We prove that the Kazhdan–Lusztig coefficient $\mu(x, w)$ lies in $\{0, 1\}$ whenever x is fully commutative and w is arbitrary in a Weyl group of type E_6 , E_7 or E_8 . Together with the known results in types A and D , this settles Green’s 0–1 question for every finite simply laced Coxeter group. The proof uses star operations to reduce to terminal (T -avoiding) upper endpoints, which we classify by exhaustive enumeration with parabolic pruning. Apart from the type- D elements classified by Gern and their parabolic embeddings, there are just two new terminals, of lengths 28 and 50. By Fan’s monomial cell theorem, each terminal has a unique eligible fully commutative lower endpoint. For each new terminal, the length gap is even, so the coefficient is zero. We also construct infinite families of full-support terminals in $\tilde{E}_8$, E_{10} and every generalized type E_{4r+1} , $r \geq 3$, for which $\mu(x, w) = 0$ for all fully commutative x . These family arguments are symbolic in their parameters, with finite computational inputs specified separately.

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1 Introduction

Let (W, S) be a Coxeter system with length function ℓ and Bruhat order $\leq$, and let $P_{x,w}(q)$ be its ordinary equal-parameter Kazhdan–Lusztig polynomials [18]. Put $\mu(x, w) = [q^{(\ell(w)-\ell(x)-1)/2}]P_{x,w}(q)$ if $x < w$ and $\ell(w) - \ell(x)$ is odd, and $\mu(x, w) = 0$ otherwise. Thus μ is the coefficient at the largest allowed degree when the gap is odd; it need not be the leading nonzero coefficient of $P_{x,w}$. A *cover* is a pair $x < w$ with $\ell(w) - \ell(x) = 1$; then $P_{x,w} = 1$ and $\mu(x, w) = 1$. Write $L(w) = \{s \in S : \ell(sw) < \ell(w)\}$ and $R(w) = \{s \in S : \ell(ws) < \ell(w)\}$ for the descent sets and $\text{supp}(w)$ for the set of generators in a reduced word of w ; w has *full support* if $\text{supp}(w) = S$. Following Stembridge [22], w is *fully commutative* (FC) if any two of its reduced words are related by interchanging adjacent commuting generators; in a simply laced system this holds if and only if no reduced word of w contains a factor sts with $m(s, t) = 3$ [22, Proposition 2.1]. The FC elements form the set W_{FC} . A set $I \subseteq S$ of pairwise commuting generators is an *independent set* of the Coxeter graph; $i(I) = \prod_{s \in I} s$ is a *commuting product*. An element is *terminal* if no reduced word for it begins or ends with two noncommuting generators, and a *noncommuting terminal* if moreover it is not a commuting product; these are Gern’s weakly bad and bad elements [10, Definition 2.1.1] and Laird’s T -avoiding and T_2 -avoiding elements [19, §2.2].

Theorem 1.1 (Finite exceptional types). *If W has type E_6 , E_7 or E_8 , then $\mu(x, w) \in \{0, 1\}$ for all $x \in W_{\text{FC}}$ and $w \in W$.*

Corollary 1.2 (Finite simply laced groups). *If W is a finite simply laced Coxeter group, $x \in W_{\text{FC}}$ and $w \in W$, then $\mu(x, w) \in \{0, 1\}$.*

The generalized diagram E_n , $n \geq 6$, is the chain $0-1-\cdots-(n-2)$ with node $n-1$ attached to node 2 (Figure 1); thus $E_9 = \tilde{E}_8$, and E_n is indefinite for $n \geq 10$. Section 4.2 defines, for every $r \geq 3$, explicit real roots $\beta_{r,k}$ ($k \geq 0$) of E_{4r+1} , with reflections $b_{r,k} = r_{\beta_{r,k}}$, and an independent set I_r of size $2r+1$.

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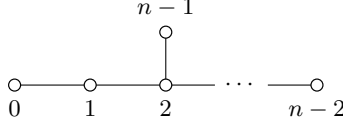


Figure 1: Vertex numbering for E_n .

Theorem 1.3 (A rank-uniform family). *For every $r \geq 3$ the reflections $b_{r,k}$, $k \geq 0$, are pairwise distinct full-support terminal elements of $W(E_{4r+1})$ that are not fully commutative, with $L(b_{r,k}) = R(b_{r,k}) = I_r$. For every fully commutative x and every $k \geq 0$, $\mu(x, b_{r,k}) = 0$.*

An analogous family occurs in $\tilde{E}_8$ with exact lengths $27 + 92k$ (Theorem 4.7), and another in E_{10} (Proposition 4.8). Thus there are infinitely many noncommuting terminals in each E_n for $n \geq 9$.

Green [14, Introduction] asked whether $\mu(x, w) \in \{0, 1\}$ for fully commutative x in a large class of Coxeter groups, “possibly all of them”; Chmutov [4, abstract and §2.3] calls this *Green’s 0–1 conjecture*. It is known in types A and $\tilde{A}$ (Green [14, Theorem 4.1; Theorem 3.1 of the arXiv version]), $B = C$ (Chmutov [4, Theorem 2.3.8]) and D (Gern [10, Theorem 4.5.11]). Theorem 1.1 completes the finite simply laced case. The finite irreducible types not covered by these results are F_4 , H_3 and H_4 ; dihedral groups, including G_2 , have $P_{x,w} = 1$ for $x \leq w$ [20, Proposition 7.3]. Finiteness cannot be dropped from Corollary 1.2: $\mu = 2$ occurs in $\tilde{D}_4$. The FC hypothesis is also essential, since $\mu = 10$ occurs in E_6 with a non-FC lower endpoint (Section 5).

Jones [17, Theorems 1.2 and 6.1] proved the bound for a Deodhar upper endpoint in every finite Weyl group (E_6 , E_7 , E_8 , F_4 and G_2 by computer). This uses a hypothesis on the upper endpoint. When both endpoints are fully commutative the bound holds in every E_n by Graham’s thesis [11], as reported in Green [13, Remark 9.5].

The reduction to terminal upper endpoints (Lemma 2.2) is Green’s argument as adapted by Gern, and the lower-endpoint lemma (Lemma 2.4) is Fan’s monomial cell theorem [6] in Green’s formulation [13]; what is new is the classification of the terminal elements of E_6 , E_7 and E_8 (Proposition 3.1) by exhaustive enumeration, with parabolic pruning in place of the 696,729,600 elements of E_8 . The noncommuting terminals are Gern’s type- D list together with exactly two new elements, $w_7^E \in E_7$ of length 28 and $w_8^E \in E_8$ of length 50 (Remark 3.4); each has a unique fully commutative lower endpoint satisfying both descent inclusions, with even length gap for the new ones, so the only Kazhdan–Lusztig value needed is Gern’s D_6 value $\mu(x_6, w_6) = 1$ [10, Lemma 4.5.5] (du Cloux’s Coxeter; recomputed in Appendix A).

For a non-cover pair with nonzero $\mu(x, w)$, both descent sets of w are contained in the corresponding descent sets of x . Star operations decrease the upper endpoint’s length until either a cover is reached or the upper endpoint is terminal. For each terminal in the finite list, the common descent set is a maximum independent set on its support. By the lower-endpoint lemma, x is the product of those descents. The length gap is even except for the inherited D_6 pair. Covers have $\mu = 1$.

For the infinite families, vanishing follows from the maximum-descent criterion and the exclusion of FC covers. We exclude covers of reflections using the lifting property and a braid in a reduced palindromic word (Lemma 4.2). For the affine family, five braid witnesses occur as prefixes in length-additive translation factorizations. The proof inputs and supplementary checks are listed in Appendix A.

Infinite families of T -avoiding elements inside a fixed infinite Coxeter group were known in affine type A with an even number of generators (Fan–Green [7, Proposition 3.1.2], fully commutative), and in affine type C (the sandwich stacks of Laird [19, §2.3, Figures 2.8–2.9], within Ernst’s type-II non-cancellable elements [5, §3.2]). For type F_n , Laird [19, §3.3] reports unpublished non-FC constructions of Cross, Ernst, Hills–Kimball and Quaranta (2012) and Gilbertson–Ernst (2013). Laird’s 2016 list of open classification problems includes finite type E_n as well as affine types $\tilde{E}_6$, $\tilde{E}_7$ and $\tilde{E}_8$ [19, §5.3, p. 63]. Recent work on star and weak star irreducibility in affine types $\tilde{B}$ and $\tilde{D}$ concerns fully commutative elements [2], whereas the families here are non-FC. By Green’s theorem [12, Theorem 6.3] every E_n is star reducible, so every fully commutative terminal in E_n is a commuting product, which is why all families here are non-FC.

Related computational work includes Geck’s *PyCox* [8] and Geck–Halls’s analysis of the cells of E_8 [9]. The cell partition does not specify all coefficients required for the bound: the FC elements are not a union of Kazhdan–Lusztig left cells in type E [15, Example 2.2.5]. Green–Losonczy attribute this phenomenon to

Graham's thesis [11, §9.9]. Moreover, the off-diagonal action in a left-cell W -graph only uses a coefficient $\tilde{\mu}(x, w)$ when some generator belongs to $L(x) \setminus L(w)$ [9, Remark 2.1]; equal-descent pairs, such as the residual terminal pairs, have no corresponding off-diagonal coefficient in this action. To our knowledge, the exceptional bound, the terminal lists of Table 1 and the infinite families have not appeared before.

Sections 2, 3 and 4 contain the structural lemmas, the finite theorem and the infinite families, Section 5 complements and open problems, and Appendix A the computational certificates.

2 Structural reductions

2.1 Star reduction

We use the elementary properties of μ from [18, §2.3 and Theorem 4.2], restated in [10, Proposition 1.2.4].

Lemma 2.1 (Descent criterion). *Let $x < w$. If $s \in L(w) \setminus L(x)$, then $\mu(x, w) = 0$ unless $x = sw$; if $s \in R(w) \setminus R(x)$, then $\mu(x, w) = 0$ unless $x = ws$.*

Proof. This is [18, (2.3.e) and (2.3.f)]; see [10, Proposition 1.2.4(3),(4)]. $\square$

For noncommuting generators s, t with $m(s, t) = 3$ let $D_L(s, t) = \{w : |L(w) \cap \{s, t\}| = 1\}$. For $w \in D_L(s, t)$ exactly one of sw, tw lies in $D_L(s, t)$; it is the left star *w . Right stars w^* are defined symmetrically. Stars preserve full commutativity in simply laced systems [12, Introduction]; see also [10, Proposition 1.1.29]. By star transport [18, Theorem 4.2], also stated in [10, Proposition 1.2.4(1),(2)], for $x, w \in D_L(s, t)$,

$$\tilde{\mu}(x, w) = \tilde{\mu}({}^*x, {}^*w), \quad (1)$$

where $\tilde{\mu}(y, z)$ equals $\mu(y, z)$ if $y < z$, $\mu(z, y)$ if $z < y$, and 0 otherwise; the same holds for right stars.

Lemma 2.2 (Terminal reduction). *Let W be simply laced. If $\mu(x, b) \in \{0, 1\}$ for every terminal upper endpoint b and every fully commutative x , then $\mu(x, w) \in \{0, 1\}$ for all $x \in W_{\text{FC}}$ and all $w \in W$.*

Proof. This is the argument of Green [14, proof of Theorem 3.1 (published Theorem 4.1)], as adapted by Gern [10, Lemma 2.3.10 and Theorem 4.5.11]. We induct on $\ell(w)$. Put $d = \ell(w) - \ell(x)$; only $x < w$ with $d \geq 3$ odd needs proof. If $L(w) \not\subseteq L(x)$ or $R(w) \not\subseteq R(x)$, the coefficient is zero by Lemma 2.1. Assume both inclusions, whence $L(w)$ and $R(w)$ are commuting, like the descent sets of the fully commutative x . If w is terminal, the hypothesis applies. Otherwise let $w = stv$ be reduced with $m(s, t) = 3$ (a noncommuting suffix is treated with right stars). Then $s \in L(w)$, $t \notin L(w)$ because $L(w)$ is commuting, and $s, t \notin L(v)$ (t by reducedness, and if $s \in L(v)$, then $w = stsv' = tstv'$ and $t \in L(w)$); so $\{tv, stv\}$ is the left $\{s, t\}$ -string in $D_L(s, t)$ and ${}^*w = sw$ has length $\ell(w) - 1$. By descent inclusion, $s \in L(x)$; since x is fully commutative, $t \notin L(x)$. Thus $x \in D_L(s, t)$ and *x is defined and fully commutative. By (1), $\mu(x, w) = \tilde{\mu}({}^*x, {}^*w)$; as $\ell({}^*x) = \ell(x) \pm 1$, the new length difference is d or $d - 2$, both positive, so the value is zero if the new endpoints are incomparable and lies in $\{0, 1\}$ by induction otherwise. $\square$

Remark 2.3 (A star step that reaches a cover). On the path 0–1–2, take

$$x = s_1, \quad w = s_1 s_0 s_2 s_1.$$

These words are reduced, with lengths 1 and 4, and $x < w$ by the subword property [3, Theorem 2.2.2]. Both left descent sets equal $\{s_1\}$, so both elements lie in $D_L(s_1, s_0)$. Their left stars are

$${}^*x = s_0 s_1, \quad {}^*w = s_0 s_2 s_1.$$

The upper length decreases by one and the lower length increases by one: the gap drops from 3 to 1. Deleting s_2 from the displayed reduced word for *w gives *x , so the new pair is a cover. By star transport, $\mu(x, w) = \mu({}^*x, {}^*w) = 1$. This also exhibits the exception in Lemma 2.1: $s_2 \in L({}^*w) \setminus L({}^*x)$, but ${}^*x = s_2 {}^*w$. The reduction can thus finish at a cover before reaching a terminal.

2.2 Maximum commuting descents

An FC element with a maximum independent set of common descents on its support equals the product of those descents. The type- D case is [10, Lemma 4.5.8]. We use Fan's monomial cell theory for the Temperley–Lieb quotient [6], in the formulation of Green [13, §4]; statement numbers refer to arXiv:0704.0283v1.

Write $b_y, y \in W_{\text{FC}}$, for the monomial basis of this quotient. Two elements lie in the same monomial right cell if each of their basis elements can be obtained from the other by right multiplication by monomial basis elements, up to a nonzero scalar. Left cells are defined using left multiplication. These are cells of the quotient's monomial basis; we do not identify them with ordinary Kazhdan–Lusztig cells. Define

$$a_{\text{TL}}(y) = \max\{|A| : y = u i(A) v \text{ length-additively, } A \text{ independent}\}.$$

This is Fan's a -function [6, Definition 2.3.1, p. 141], as in [13, Definition 4.1]. We use two facts:

1. The monomial right cell of $i(A)$ consists exactly of the FC elements $y = i(A)v$ with length additive and $a_{\text{TL}}(y) = |A|$. The analogous statement for left cells uses a suffix $i(A)$ [13, Proposition 4.4(iv)], [6, Proposition 4.4.3 and its dual, p. 148].
2. A monomial left cell and a monomial right cell in the same two-sided cell intersect in exactly one element [13, Proposition 4.4(v)], [6, p. 140 and proof of Theorem 6.1.2, p. 154].

These facts hold throughout the series $E_n, n \geq 6$, including the affine and indefinite cases.

Lemma 2.4 (Maximum common commuting descents). *Let $n \geq 6$, $W = W(E_n)$, and let $x \in W_{\text{FC}}$ with $\text{supp}(x) \subseteq K$. Let I be an independent set of maximum cardinality in the subgraph induced by K . If $I \subseteq L(x) \cap R(x)$, then $x = i(I)$.*

Proof. Because the generators in I commute and are left descents, $i(I)$ is a reduced prefix of x . Since they are also right descents, $i(I)$ is a reduced suffix; the two occurrences can overlap. Hence $a_{\text{TL}}(x) \geq |I|$. Every commuting factor in a reduced expression for x uses an independent subset of K . Since I has maximum cardinality, $a_{\text{TL}}(x) \leq |I|$.

Thus $a_{\text{TL}}(x) = |I|$. By Fact (1), x belongs to both the monomial right cell and the monomial left cell of $i(I)$. Their intersection contains $i(I)$ and has exactly one element by Fact (2). Therefore $x = i(I)$. This is the singleton-cell argument used at the end of [13, proof of Lemma 5.2]. $\square$

The word *maximum* is essential. On a three-node path 0–1–2, the set $\{1\}$ is maximal under inclusion, but the independent set $\{0, 2\}$ is larger. The FC element $s_1 s_0 s_2 s_1$ has both descent sets equal to $\{s_1\}$ and is not s_1 . For a maximal independent set, the upper bound on $a_{\text{TL}}(x)$ can therefore fail.

Proposition 2.5 (Maximum-descent parity criterion). *Let $n \geq 6$ and let $b \in W(E_n)$ satisfy $L(b) = R(b) = I$, where I is a maximum independent set in the subgraph induced by $\text{supp}(b)$. If $\ell(b) \equiv |I| \pmod{2}$, then for every $x \in W_{\text{FC}}$,*

$$\mu(x, b) \in \{0, 1\}, \quad x < b, \ell(b) - \ell(x) > 1 \implies \mu(x, b) = 0.$$

Proof. The type- D prototype is [10, Lemmas 4.5.4 and 4.5.9]. Let $x < b$ be fully commutative with $\ell(b) - \ell(x) > 1$ and $\mu(x, b) \neq 0$. By Lemma 2.1, $I \subseteq L(x) \cap R(x)$, and by the subword property, $\text{supp}(x) \subseteq \text{supp}(b)$. By Lemma 2.4 with $K = \text{supp}(b)$, $x = i(I)$. Then $\ell(b) - \ell(x) = \ell(b) - |I|$ is even and $\mu(x, b) = 0$, a contradiction. Covers have $\mu = 1$. $\square$

Remark 2.6. Proposition 2.5 does not require b to be terminal. The parity hypothesis cannot be dropped: Section 5.3 exhibits elements of $\tilde{E}_8$ satisfying every other hypothesis with $\mu(i(I), b) = 1$.

Remark 2.7. The independence number of E_n is $\lceil n/2 \rceil$. Among independent sets excluding node 2, the maximum size is $2 + \lceil (n-4)/2 \rceil$; among those including it, the maximum is $2 + \lceil (n-5)/2 \rceil$. The larger of the two is $\lceil n/2 \rceil$.

2.3 Geometric terminal test

Normalize simple roots by $(\alpha_j, \alpha_j) = 2$ and $(\alpha_s, \alpha_t) = -1$ for adjacent nodes, and write $s \sim t$ when s and t are adjacent, i.e. $m(s, t) = 3$. For a real root β the reflection r_β acts by $r_\beta(\rho) = \rho - (\rho, \beta)\beta$; W acts on the left and the matrix of w has columns $w(\alpha_j)$. Root inequalities refer to the simple-root basis, and $s \in R(w)$ if and only if $w(\alpha_s) < 0$ [3, Proposition 4.4.6]. Since $(ws)(\alpha_t) = w(\alpha_s + \alpha_t)$ for $s \sim t$, the right-terminal test is

$$s \in R(w) \implies w(\alpha_s + \alpha_t) > 0 \quad \text{for every } t \sim s, \quad (2)$$

and testing w^{-1} gives the left-terminal condition.

3 Finite exceptional types

3.1 The terminal classification

A fully commutative x is *eligible* for a terminal b if $x \leq b$, $L(b) \subseteq L(x)$ and $R(b) \subseteq R(x)$; by Lemma 2.1, every ineligible non-cover has $\mu(x, b) = 0$. Elements are written as strings of node labels in the numbering of Section 1: the chain $0-1-\dots-(n-2)$ with node $n-1$ attached to node 2.¹ A string denotes the product of the corresponding simple reflections in the order written, with the left action and column convention of Section 2.3. Every element displayed in Table 1 and in the affine family of Section 4.3 is an involution, so a string and its reverse denote the same element.

Proposition 3.1 (Computational classification). *The noncommuting terminals in E_6 , E_7 and E_8 are exactly the eleven elements in Table 1. Both descent sets of each of them equal the support of the commuting product x shown in the table. The remaining terminals are commuting products: 22, 36 and 58 of them, respectively, including the identity.*

Extending Gern's notation [10, Theorem 2.3.6], we write w_4 and w_6 for his elements of lengths 7 and 15 (viewed inside E_n as in Remark 3.4), and w_7^E , w_8^E for the two new elements of lengths 28 and 50, where w_8^E is the concatenation of the two strings

$$(7534231270123456210321432)(5437210321432543721324357).$$

Type	Terminal b	$\ell(b)$	$x = i(I)$	$\ell(b) - \ell(x)$	Source
E_6	1325213	7	135	4	w_4
E_7	1326213	7	136	4	w_4
E_7	13256213	8	1356	4	$w_4 s_5$
E_7	132543621324356	15	1356	11	w_6
E_7	1325436210321432543621324356	28	1356	24	w_7^E
E_8	1327213	7	137	4	w_4
E_8	13257213	8	1357	4	$w_4 s_5$
E_8	61327213	8	1367	4	$w_4 s_6$
E_8	132543721324357	15	1357	11	w_6
E_8	1325437210321432543721324357	28	1357	24	w_7^E
E_8	w_8^E (displayed above)	50	1357	46	w_8^E

Table 1: The noncommuting terminals of E_6 , E_7 , E_8 : a reduced word, the length, the unique eligible fully commutative lower endpoint (Corollary 3.2), the gap, and the identification of Remark 3.4. Only the rows of length 15, Gern's D_6 pair, have odd gap.

Corollary 3.2. *Each terminal in Table 1 has the unique eligible fully commutative lower endpoint x displayed there.*

¹In Bourbaki's labelling of E_n (chain $1-3-4-\dots-n$, node 2 attached to node 4), our node 0 is Bourbaki's node 1, our node j with $1 \leq j \leq n-2$ is Bourbaki's node $j+2$, and our node $n-1$ is Bourbaki's node 2.

Proof of Corollary 3.2. The induced support graphs have types D_4 , $D_4 \sqcup A_1$, D_6 , E_7 and E_8 , with independence numbers 3, 4, 4, 4, 4: the matchings (1, 2) in the rows of lengths 7 and 8, (1, 2), (3, 4) in the rows of length 15, (0, 1), (2, 3), (4, 5) in the rows of length 28 and (0, 1), (3, 4), (5, 6), (2, 7) in the row of length 50 bound every independent set by the number of support nodes minus the matching size, and the displayed descent set attains the bound (compare Remark 2.7). If y is eligible, then $\text{supp}(y) \subseteq \text{supp}(b)$, so $y = x$ by Lemma 2.4; conversely x , a commuting product of left descents of b , is a reduced prefix of b , hence below b and eligible. $\square$

3.2 Exhaustiveness, and the proofs of Theorem 1.1 and Corollary 1.2

All computations use the faithful integral geometric representation [3, Theorem 4.2.7]: right multiplication by s_i replaces the columns $a_j = w(\alpha_j)$ by $a_i \mapsto -a_i$ and $a_j \mapsto a_j + a_i$ for $j \sim i$. The programs compute descents from column signs and test terminality on both sides using (2). In E_6 and E_7 , breadth-first closure under all generators enumerates the whole group. For E_8 , fix $J \subseteq S$ and write $w = av$ length-additively with $a \in W^J$ (minimal coset representatives) and $v \in W_J$ [3, Proposition 2.4.4 and Corollary 2.4.5]; call w *right-terminal* if no reduced word of w ends in a noncommuting pair.

Lemma 3.3 (Parabolic pruning). *If $w = av$ is right-terminal, then v is right-terminal in W_J .*

Proof. A reduced word for v ending in adjacent generators, prefixed by a reduced word for a , gives such a word for w , by length additivity. $\square$

The enumeration takes as input the finite Coxeter diagram and a chain $\emptyset = K_0 \subset K_1 \subset \dots \subset K_m = S$ of standard parabolic subsets; its output is the complete terminal set. Write $\text{RT}_K(u)$ for the right-terminal test (2) using the generators and edges in K .

Parabolic terminal enumeration

Initialize $\mathcal{R}_0 = \{e\}$.

For $i = 1, \dots, m$:

Set $K = K_i$, $J = K_{i-1}$.

Enumerate $C_i = \{a \in W_K : R(a) \cap J = \emptyset\}$.

Set $\mathcal{R}_i = \{av : a \in C_i, v \in \mathcal{R}_{i-1}, \text{RT}_K(av)\}$.

Return $\{w \in \mathcal{R}_m : \text{RT}_S(w^{-1})\}$.

To enumerate C_i , start a queue with e . For each queued a and $s \in K$, strip right J -descents from sa until none remains, and enqueue the resulting element if new. Stripping strictly decreases length and gives the unique minimal representative of saW_J . Left multiplication acts transitively on W_K/W_J , so an exhausted queue contains every coset representative. The implementation also checks $|C_i| = [W_K : W_J]$.

The invariant is that $\mathcal{R}_i$ contains exactly the right-terminal elements of W_{K_i} . Every retained product passes the defining test; conversely, the unique length-additive parabolic factorization $w = av$ of any right-terminal w has $v \in \mathcal{R}_{i-1}$ by Lemma 3.3 and induction. Thus it is tested and retained. The program compares elements by their matrices in the faithful representation. Each queue is finite because W_K is finite, and the procedure stops after the last parabolic stage; there is no imposed word-length cutoff. The inverse test then imposes left terminality.

The implementation uses the chain $E_8 \supset E_7 \supset E_6 \supset D_5 \supset D_4 \supset A_3 \supset A_2 \supset A_1 \supset \{e\}$, traversed from smallest to largest. The final E_7 parabolic has 576 right-terminals and 240 cosets, so this stage tests 138,240 products (143,660 over all stages). Of these, 2,160 are right-terminal in E_8 , and the inverse test leaves exactly the 64 terminals of Proposition 3.1. For the finite theorem, it remains to check the eligible pairs of Corollary 3.2.

Proof of Theorem 1.1. By Lemma 2.2 it suffices to treat terminal b . If b is a commuting product, every $x < b$ is a proper subproduct, so some $s \in L(b) \setminus L(x)$ exists, and $\mu(x, b) \in \{0, 1\}$ by Lemma 2.1. For a noncommuting terminal, there is one eligible non-cover pair (x, b) per row of Table 1, by Corollary 3.2 and Lemma 2.1. The length gap is even, and hence the coefficient is zero, in every row except the two of length 15. These are the same pair in a D_6 parabolic subgroup: in E_n , $n = 7, 8$, mapping Gern's labels

$1, \dots, 6$ to $1, n-1, 2, 3, 4, 5$ identifies it with Gern's $x_6 = (-1, -2, 4, 3, 6, 5)$ and $w_6 = (-1, -6, 3, -4, 5, -2)$ in signed permutation notation [10, Example 1.1.6]. Bruhat order and Kazhdan–Lusztig polynomials of a standard parabolic subgroup are the restrictions of those of W , since reduced words of W_J are reduced in W and the Hecke algebra of W_J is a bar-stable subalgebra containing its Kazhdan–Lusztig basis [20, §9.6 and Lemma 9.10(c),(d), with $z = e$]. Gern computed $\mu(x_6, w_6) = 1$ with du Cloux's Coxeter [10, Lemma 4.5.5] (recomputed in Appendix A). Hence every coefficient with terminal upper endpoint lies in $\{0, 1\}$. $\square$

Proof of Corollary 1.2. The irreducible finite simply laced types are A , D and E ; combine Green's theorem, Gern's theorem and Theorem 1.1. For a direct product $W = \prod_j W_j$, full commutativity is componentwise and $P_{x,w} = \prod_j P_{x_j, w_j}$: the Hecke algebra and its bar involution factor as tensor products, and the tensor product of the component Kazhdan–Lusztig bases satisfies the defining invariance and triangularity conditions [3, §5.1]. Let $x < w$, let d_j be the component length differences, $d = \sum_j d_j$, and m the number of j with $d_j > 0$. For even d , $\mu = 0$; for odd d and $m \geq 2$, since $\deg P_{x_j, w_j} \leq (d_j - 1)/2$ for $d_j > 0$, we have $\deg P_{x,w} \leq (d - m)/2 < (d - 1)/2$, so $\mu = 0$; for $m = 1$, $\mu(x, w)$ is the coefficient in the single nontrivial component, which lies in $\{0, 1\}$. $\square$

3.3 Relation to the type- D list and further structure

Remark 3.4 (Relation to Gern's classification). The parabolic subgroup of E_n on $\{1, n-1, 2, \dots, n-2\}$ has type D_{n-1} ; sending Gern's labels $1, 2, j$ ($3 \leq j \leq 7$) to $1, n-1, j-1$ identifies it with Gern's $W(D_{n-1})$, and then the rows of Table 1 of lengths 7, 8 and 15 are Gern's bad elements w_4 , $w_4 u$ and w_6 of [10, Theorem 2.3.6], while the E_8 row of length 28 is w_7^E in the parabolic subgroup on $\{0, \dots, 5, 7\}$. Thus the list has the form bu , where $b \in \{w_4, w_6, w_7^E, w_8^E\}$ belongs to the ambient group and u is a commuting product of generators neither in nor adjacent to $\text{supp}(b)$. In E_6 only w_4 occurs; in E_7 the possibilities are w_4, w_6, w_7^E ; in E_8 all four occur. The unique longest terminals in E_7 and E_8 are w_7^E and w_8^E , respectively.

Remark 3.5 (Additional structure). The terminal-structure certificate records reduced layered words and orthogonal-reflection decompositions. The element w_4 is a reflection, w_6 is a product of three orthogonal reflections, and w_7^E and w_8^E are products of four and two pairwise orthogonal reflections, respectively. In the compatible embeddings into E_8 they form a right weak order chain $w_4 < w_6 < w_7^E < w_8^E$. These observations are not needed for Theorem 1.1.

For compact descriptions of the two new terminals, in the respective E_7 and E_8 simple-root coordinates put

$$\begin{aligned} (\rho_1, \rho_2, \rho_3, \rho_4) &= ((0, 0, 1, 1, 1, 1, 1), (0, 1, 1, 1, 1, 0, 1), \\ &\quad (0, 1, 2, 1, 0, 0, 1), (1, 2, 2, 2, 1, 1, 1)), \\ \eta_1 &= (1, 2, 3, 3, 2, 2, 1, 2), \quad \eta_2 = (1, 3, 4, 3, 2, 1, 0, 2). \end{aligned}$$

These are positive roots, pairwise orthogonal within each set, and

$$w_7^E = r_{\rho_1} r_{\rho_2} r_{\rho_3} r_{\rho_4}, \quad w_8^E = r_{\eta_1} r_{\eta_2}.$$

The verification program compares these products with the reduced words in Table 1 and records the results in the certificate.

For $b = w_7^E, w_8^E$, an alternative proof of $\mu(i(I), b) = 0$ uses Lusztig's a -function: $J = I \cup \{0\}$ has type $A_2 \times A_1^3$ and $\ell(w_0(J)) = 6$, and $b = u w_0(J) v$ length-additively with lengths $11 + 6 + 11$ and $11 + 6 + 33$ (words in the certificate). A length-increasing simple multiplication decreases the Kazhdan–Lusztig left or right preorder [9, §2], so $b \leq_{LR} w_0(J)$. Consequently Lusztig's a -function satisfies $a(b) \geq a(w_0(J)) = 6$, by [20, Proposition 13.8 and properties P4, P12 of §14.2, proved in Chapter 15]. Since $I \subseteq L(b)$, we have $P_{i(I), b} = P_{e, b}$ by repeated left descent invariance [3, Proposition 5.1.8 and its left analogue]. In the equal-parameter normalization, $\Delta(b) = \ell(b) - 2 \deg P_{e, b}$; by property P1, $a(b) \leq \Delta(b)$ [20, §14.1 and §15.2]. Thus $\deg P_{i(I), b} \leq (\ell(b) - 6)/2$, below the degree $(\ell(b) - 5)/2$ of $\mu(i(I), b)$. Likewise $a(w_6) \geq 5$, consistent with $a(w_6) = 5$ [10, Proposition 4.5.1].

4 Infinite families

For the infinite families, coefficients at non-covers vanish by the maximum-descent parity criterion. Since covers have coefficient 1, we must also exclude fully commutative covers. For reflections, we use a reduced palindromic word; for the affine family, we use five braid witnesses and a length-additive translation factorization.

4.1 Reflections and their covers

The *depth* $\text{dp}(\beta)$ of a positive root β is the least d such that $\beta = s_1 \cdots s_{d-1} \alpha_t$ for simple reflections s_i and a simple root α_t (cf. [3, §4.6]).

Lemma 4.1 (Reflections). *Let (W, S) be simply laced, let β be a positive root and $d = \text{dp}(\beta)$.*

- (i) $\ell(r_\beta) = 2d - 1$, and for every expression $\beta = s_1 \cdots s_{d-1} \alpha_t$ of minimal length the palindromic word $s_1 \cdots s_{d-1} t s_{d-1} \cdots s_1$ is a reduced word for r_β .
- (ii) If $s \in S$ and $(\beta, \alpha_s) < 0$, then $s\beta$ is a positive root of depth $d+1$, $\ell(sr_\beta) = \ell(r_\beta) + 1$ and $\ell(sr_\beta s) = \ell(r_\beta) + 2$. If $(\beta, \alpha_s) > 0$ and $\beta \neq \alpha_s$, then $s\beta$ is a positive root of depth $d - 1$.
- (iii) r_β is fully commutative if and only if β is simple.

Proof. Recall that $\ell(ws) > \ell(w)$ if and only if $w(\alpha_s) > 0$, and that s negates only $\pm\alpha_s$. Let $(\beta, \alpha_s) = -c < 0$. Then $s\beta = \beta + c\alpha_s > 0$; since $r_\beta(\alpha_s) = \alpha_s + c\beta > 0$, $s \notin R(r_\beta) = L(r_\beta)$, so $\ell(sr_\beta) = \ell(r_\beta) + 1$; and since $(sr_\beta)(\alpha_s) = -\alpha_s + c s\beta = c\beta + (c^2 - 1)\alpha_s > 0$, $\ell(sr_\beta s) = \ell(r_\beta) + 2$. As $sr_\beta s = r_{s\beta}$, this shows $\ell(r_{s\beta}) = \ell(r_\beta) + 2$ when $(\beta, \alpha_s) < 0$ and, applied to $s\beta$, $\ell(r_{s\beta}) = \ell(r_\beta) - 2$ when $(\beta, \alpha_s) > 0$, $\beta \neq \alpha_s$. Now $\ell(r_\beta) = 2 \text{dp}(\beta) - 1$ by induction on the odd number $\ell(r_\beta)$ (the case $\ell(r_\beta) = 1$ is immediate): if $\ell(r_\beta) \geq 3$, some s has $(\beta, \alpha_s) > 0$, $\beta \neq \alpha_s$ (as $(\beta, \beta) = 2$, $\beta \geq 0$), so $\beta' = s\beta > 0$ has $\ell(r_{\beta'}) = \ell(r_\beta) - 2 = 2 \text{dp}(\beta') - 1$ and $\text{dp}(\beta) \leq \text{dp}(\beta') + 1$, hence $2 \text{dp}(\beta) - 1 \leq \ell(r_\beta)$. From the palindromic expression, $\ell(r_\beta) \leq 2d - 1$. This proves (i); (ii) follows from (i) and the two length formulas. For (iii), if $d \geq 2$ the reduced word of (i) contains $s_{d-1} t s_{d-1}$, which equals t unless $m(s_{d-1}, t) = 3$; so it is a braid factor and $r_\beta \notin W_{\text{FC}}$ [22, Proposition 2.1]. $\square$

Lemma 4.2 (Covers of terminal reflections). *Let $n \geq 6$, $W = W(E_n)$, and let $b = r_\beta$ for a positive root β with at least three nonzero coordinates. Suppose $L(b) = R(b) = I$, where I is a maximum independent set of the subgraph induced by $\text{supp}(b)$ and $|I|$ is odd. Then no Bruhat cover of b is fully commutative. Consequently $\mu(x, b) = 0$ for every fully commutative x .*

Proof. Suppose, for a contradiction, that x is a fully commutative cover of b . If $I \subseteq L(x) \cap R(x)$, then $\text{supp}(x) \subseteq \text{supp}(b)$, so by Lemma 2.4, $x = i(I)$ has odd length $|I|$, impossible as $\ell(b)$ is odd. Otherwise some $s \in I$ has $s \notin L(x)$ (or, symmetrically, $s \notin R(x)$); by the lifting property [3, Proposition 2.2.7], $x \leq sb$, hence $x = sb = (bs)^{-1}$ by length and involutivity, and it suffices to show $bs \notin W_{\text{FC}}$. Put $m = (\alpha_s, \beta)$. Since $s \in R(b)$, $b(\alpha_s) = \alpha_s - m\beta < 0$, so $m \geq 1$, and $s\beta = \beta - m\alpha_s$ is a positive non-simple root, its coordinates off s being those of β . Since $sbs = r_{s\beta}$ and $(\alpha_s, s\beta) = -m < 0$, we have $\ell(sr_{s\beta}) = \ell(r_{s\beta}) + 1$ by Lemma 4.1(ii), so s followed by the palindromic word of Lemma 4.1(i) for $r_{s\beta}$ is a reduced word for bs containing a braid factor; hence $bs \notin W_{\text{FC}}$. For non-covers, the coefficient is zero by Proposition 2.5, since $\ell(b)$ and $|I|$ are odd. $\square$

4.2 The rank-uniform construction

To apply Lemma 4.2, we construct a positive real root β with full support whose positive simple-root pairings occur on a maximum independent set I of odd size. With $m_j = (\alpha_j, \beta)$,

$$r_\beta(\alpha_j) = \alpha_j - m_j \beta.$$

Since β has full support, this root is negative precisely when $m_j > 0$. For an edge $s \sim t$ incident to I with $m_s + m_t \leq 0$,

$$r_\beta(\alpha_s + \alpha_t) = \alpha_s + \alpha_t - (m_s + m_t)\beta > 0,$$

as required by the terminal test.

The seed has alternating pairings along the long arm. We prove that it is a real root, then compute its orbit under a product of two reflections in an embedded affine subsystem. The orbit is quadratic in the parameter, and the pairings satisfy these inequalities throughout. In an indefinite root system, a vector of norm 2 need not be a real root.

Fix an integer $r \geq 3$, put $n = 4r + 1$, and use the E_n numbering of Section 1, so node $4r$ is attached to node 2. Define β_r by its coordinates $(\beta_r)_0 = r - 1$, $(\beta_r)_1 = r$, $(\beta_r)_2 = 2r - 1$, $(\beta_r)_{4r} = r$ and $(\beta_r)_j = 2r - \lfloor j/2 \rfloor$ for $3 \leq j \leq 4r - 1$, where $(v)_j$ denotes the coefficient of α_j in v ; so the long-arm coordinates are $2r - 1, 2r - 2, 2r - 2, \dots, 1, 1$. Put $I_r = \{0, 3, 5, \dots, 4r - 1, 4r\}$ and $a = 2r - 4$. On the embedded affine E_8 subdiagram $\{0, \dots, 7, 4r\}$ set $(\delta_0, \dots, \delta_7) = (2, 4, 6, 5, 4, 3, 2, 1)$, $\delta_{4r} = 3$, $(\gamma_0, \dots, \gamma_7) = (1, 2, 3, 2, 2, 1, 1, 0)$, $\gamma_{4r} = 1$, with all other coordinates zero. Set $T = r_\gamma r_{\gamma+\delta}$ and, for $k \geq 0$,

$$\beta_{r,k} = \beta_r - ak\gamma + (ak^2 + rk)\delta, \quad b_{r,k} = r_{\beta_{r,k}}. \quad (3)$$

Figure 2 shows the affine subdiagram and the independent set in the first case, $r = 3$.

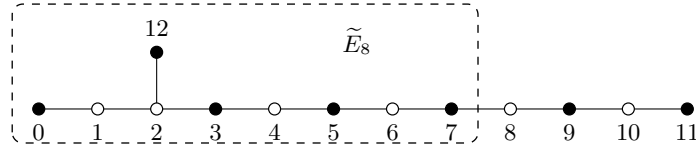


Figure 2: The affine subsystem $\{0, \dots, 7, 12\}$ in E_{13} (dashed outline) and $I_3 = \{0, 3, 5, 7, 9, 11, 12\}$ (filled vertices).

Lemma 4.3. *For every $r \geq 3$, β_r is a real root of E_{4r+1} .*

Proof. Deleting node 0 leaves finite type D_{4r} ; realize its simple roots in an orthonormal basis $e_1, \dots, e_{4r}$ as $\alpha_1 = e_1 + e_2$, $\alpha_{4r} = e_2 - e_1$, $\alpha_j = e_{j+1} - e_j$ ($2 \leq j \leq 4r - 1$), take z orthogonal to them with $(z, z) = 2 - r$, and put $\alpha_0 = z - \frac{1}{2} \sum_i e_i$; these vectors have the Cartan inner products of E_{4r+1} and form a basis, the Cartan determinant $9 - n$ being nonzero. Then $\beta_r = (r - 1)z + \sum_i c_i e_i$ with $c_i = -(r - 1)/2$ for odd and $(3 - r)/2$ for even i . Since $(\beta_r, \alpha_0) = r - 2$, $s_0 \beta_r = F \alpha_0$, where F flips the $2r$ even coordinates. As $F = \prod_{j=1}^r r_{e_{4j-2} - e_{4j}} r_{e_{4j-2} + e_{4j}} \in W(D_{4r})$ [10, Example 1.1.6], $\beta_r = s_0 F \alpha_0$ is a real root. $\square$

Lemma 4.4. *Suppose γ and $\gamma + \delta$ are real roots with $(\gamma, \gamma) = 2$ and $(\gamma, \delta) = (\delta, \delta) = 0$, and let $T = r_\gamma r_{\gamma+\delta}$. Then*

$$(T - 1)\rho = (\rho, \delta)\gamma - ((\rho, \gamma) + (\rho, \delta))\delta, \quad (T - 1)^3 = 0,$$

and if $(\beta, \delta) = -a$ and $(\beta, \gamma) = -c$, then

$$T^k \beta = \beta - ak\gamma + (ak^2 + ck)\delta \quad (k \geq 0). \quad (4)$$

Proof. Compose the two reflection formulas. Put $N = T - 1$. Since $N\delta = 0$ and $N\gamma = -2\delta$, we have $N^3 = 0$. Also $N\beta = -a\gamma + (a + c)\delta$, $N^2\beta = 2a\delta$; expand $(1 + N)^k$ binomially. The lemma holds on the whole ambient root space, where δ need not be in the radical. $\square$

Lemma 4.5. *For all $r \geq 3$ and $k \geq 0$, $\beta_{r,k} = T^k \beta_r$ is a real root with every coordinate strictly positive. Its reflection $b_{r,k}$ is terminal and has both descent sets equal to I_r .*

Proof. The vector γ is a root of the finite E_8 parabolic subsystem obtained by omitting node 7, and $\gamma + \delta$ is an affine real root (explicit lowering sequences are given in Appendix A.3). As $(\beta_r, \gamma) = -r$ and $(\beta_r, \delta) = -a$, we have $T^k \beta_r = \beta_{r,k}$ by (4). This is a real root, and the binomial increments $(a + r)\delta - a\gamma$ and $2a\delta$ are coordinatewise nonnegative for $r \geq 3$, so every coordinate of $\beta_{r,k}$ is strictly positive. The pairings $m_j = (\alpha_j, \beta_{r,k})$ are

$$\begin{aligned} m_0 &= r - 2, & m_1 &= 2 - r, & m_2 &= -1 - ak, & m_8 &= -1 - ak^2 - rk, & m_{4r} &= 1 + ak, \\ m_j &= (-1)^{j+1}(1 + ak) & (3 \leq j \leq 7), & & m_j &= (-1)^{j+1} & (9 \leq j \leq 4r - 1), \end{aligned}$$

positive exactly on I_r . Since $\beta_{r,k} > 0$, $b_{r,k}(\alpha_j) = \alpha_j - m_j \beta_{r,k}$ is negative exactly when $m_j > 0$, so $R(b_{r,k}) = I_r = L(b_{r,k})$ by involutivity. For every edge $s \sim t$ with $s \in I_r$ the sum $m_s + m_t$ is nonpositive (at $7-8$ it is $-ak(k-1) - rk$, at $8-9$ it is $-ak^2 - rk$, elsewhere by inspection), so $b_{r,k}(\alpha_s + \alpha_t) = \alpha_s + \alpha_t - (m_s + m_t)\beta_{r,k} > 0$, which is (2); the left test follows by involutivity. $\square$

For $r = 3$, we have $n = 13$, with node 12 attached to node 2. At $k = 0$ the seed and its simple-root pairings are

$$\begin{aligned}\beta_{3,0} &= (2, 3, 5, 5, 4, 4, 3, 3, 2, 2, 1, 1, 3), \\ (m_0, \dots, m_{12}) &= (1, -1, -1, 1, -1, 1, -1, 1, -1, 1, -1, 1, 1).\end{aligned}$$

At nodes 2 and 12, $m_2 = 2 \cdot 5 - 3 - 5 - 3 = -1$, while $m_{12} = 2 \cdot 3 - 5 = 1$. The positive entries occur on $I_3 = \{0, 3, 5, 7, 9, 11, 12\}$, and every edge incident to I_3 has pairing sum zero. Along the orbit, the two exceptional sums become $m_7 + m_8 = -2k(k-1) - 3k$ and $m_8 + m_9 = -2k^2 - 3k$. Both are nonpositive for every integer $k \geq 0$.

Proof of Theorem 1.3. The set I_r is independent of size $2r+1$, and the matching $(0, 1), (2, 4r), (3, 4), (5, 6), \dots, (4r-3, 4r-2)$ with $2r$ edges leaves one vertex unmatched, so I_r is a maximum independent set of E_{4r+1} (compare Remark 2.7). By Lemma 4.5, $b_{r,k}$ is a terminal reflection with $L = R = I_r$, $|I_r|$ is odd, and $\beta_{r,k}$ has $4r+1 \geq 3$ nonzero coordinates. By Lemma 4.2, $\mu(x, b_{r,k}) = 0$ for every fully commutative x , and $b_{r,k} \notin W_{\text{FC}}$ by Lemma 4.1(iii). Since $m_0 = r-2 > 0$ and $\beta_{r,k}$ is positive in every coordinate, column 0 of $b_{r,k}$ changes every coordinate row, whereas omitting a generator would leave its row unchanged; so $b_{r,k}$ has full support. The coordinate $(\beta_{r,k})_0 = r-1+4k+(4r-8)k^2$ is strictly increasing in k . Thus the positive roots, and hence their reflections, are pairwise distinct. $\square$

Remark 4.6. At $r = 3$ one has $\ell(b_{3,0}) = 75$, while the maximum lengths of fully commutative elements of $E_{10}, \dots, E_{13}$ are 55, 66, 78 and 92 (Appendix A), so a length-based exclusion of fully commutative covers already fails in rank 13; the lengths $\ell(b_{r,k}) = 8r^2 + 3 + 116k$ hold for every pair in the certificate ($r \leq 8$, $k \leq 3$; the supplement's optional extended run adds $r = 9, 10, 12$) but are not used in the proof. At $r = 2$, $\beta_2 = \delta - \gamma$, $(\alpha_0, \beta_2) = 0$, and the descent set has even size 4.

4.3 The affine family

Use E_9 numbering: chain $0-1-\dots-7$ with node 8 attached to 2; the affine node is 7. Put

$$\delta' = (2, 4, 6, 5, 4, 3, 2, 1, 3), \quad \gamma' = (1, 1, 1, 1, 1, 0, 0, 0, 0), \quad I' = \{1, 3, 5, 7, 8\},$$

and

$$c_0 = 312875645234123012856745231,$$

a reduced word for an involution. Let $T' = r_{\gamma'} r_{\delta'}$, so that $T'(\alpha) = \alpha - (\alpha, \gamma')\delta'$, and define $c_k = T'^k c_0 T'^{-k}$ for $k \geq 0$.

Theorem 4.7 (Explicit affine family). *Every c_k is a full-support terminal element of $W(\tilde{E}_8)$ that is not fully commutative, with*

$$L(c_k) = R(c_k) = I', \quad \ell(c_k) = 27 + 92k.$$

For every fully commutative $x \in W(\tilde{E}_8)$ and every $k \geq 0$, $\mu(x, c_k) = 0$.

Proof. The vector δ' is the null root and γ' is a root of the finite E_8 parabolic subsystem obtained by omitting node 7. Multiplying out the displayed word gives $c_0^2 = 1$. The columns $c_0(\alpha_j)$ of the matrix of c_0 recorded in the affine certificate (Appendix A; the certificate stores the matrix by rows) are negative precisely at the indices in I' , and for every edge incident to I' the sum of the two columns is a positive root; by (2) and involutivity, c_0 is terminal on both sides. Put $d = (2, -1, 1, -1, 1, -1, 1, -1, -1)$. Then $(\alpha_j, \gamma' - c_0 \gamma') = d_j$, and conjugation gives

$$c_k(\alpha_j) = c_0(\alpha_j) + k d_j \delta'. \quad (5)$$

As d is negative on I' and positive off I' , like the base columns, both descent sets stay I' ; on every edge $s \sim t$ with $s \in I'$ one has $d_s + d_t \geq 0$, so (5) preserves positivity of $c_k(\alpha_s + \alpha_t)$, and c_k is terminal. Full support

holds at $k = 0$ by the displayed word and at $k \geq 1$ because each descent column is strictly negative in every coordinate, whereas omitting a generator would leave its row unchanged. The c_k are pairwise distinct by (5), since every $d_j \neq 0$.

For the length, use the affine-root realization [16, §§4.1–4.3 and §6.5] and count positive roots sent negative [3, Proposition 4.4.4]: for each of the 240 finite roots η write $c_0\eta = \eta' + h\delta'$, let $\epsilon(\eta)$ be 0 for positive and 1 for negative η ; the positive affine roots $\eta + m\delta'$, $m \geq \epsilon(\eta)$, contribute $\max(0, A_\eta + kB_\eta)$ to $\ell(c_k)$, where $A_\eta = \epsilon(\eta') - h - \epsilon(\eta)$ and $B_\eta = -\sum_j \eta_j d_j$. The nonzero contributions, grouped by (A_η, B_η) , are

(A, B)	$(0, 1)$	$(0, 2)$	$(1, 0)$	$(1, 1)$	$(1, 2)$
Multiplicity	44	8	1	20	6

and the remaining 161 vanish for all $k \geq 0$; the sum is $27 + 92k$.

By Remark 2.7, I' is a maximum independent set, of size 5. Since $\ell(c_k)$ is odd, $\mu(x, c_k) = 0$ for every fully commutative non-cover x by Proposition 2.5. Let x be a fully commutative cover of c_k . If $I' \subseteq L(x) \cap R(x)$, then by Lemma 2.4, $x = i(I')$ has length $5 < \ell(c_k) - 1$. Otherwise, by the lifting property [3, Proposition 2.2.7], $x = sc_k$ or $x = c_k s$ for some $s \in I'$, as in the proof of Lemma 4.2. To exclude these possibilities for all k , write C for the matrix of c_0 , regard δ' as a column and d as a row, and put $R = 1 + \delta'd$. Since $d\delta' = 0$ and $C\delta' = \delta'$, we have by (5)

$$R^k = 1 + k\delta'd, \quad c_k = CR^k.$$

In particular $R = Cc_1 \in W$, since $C^2 = 1$. Its finite part is the identity, so the inversion calculation above has $A_\eta = 0$ and the same B_η for R^k . Thus

$$\ell(R^k) = k \sum_{\eta} \max(0, B_\eta) = 92k,$$

and $c_k = c_0 R^k$ is length-additive. For every $s \in I'$, $sc_k = (sc_0)R^k$ is also length-additive, since $\ell(sc_k) = 26 + 92k = \ell(sc_0) + \ell(R^k)$. The following reduced words for sc_0 each have length 26; brackets mark a noncommuting braid and are not part of the word:

s	Reduced word for sc_0
1	3[282]7564534123012856745231
3	1[282]7564534123012856745231
5	31[282]764534123012856745231
7	31[282]564534123012856745231
8	3[121]8756453423012856745231

Their products and reducedness are checked by the simple-root action in Appendix A. Each braid persists in a reduced word for sc_k , so sc_k is not FC; inversion gives the same conclusion for $c_k s$. Hence $\mu(x, c_k) = 0$ for every fully commutative x . Finally, $c_k \notin W_{\text{FC}}$, since otherwise we would have $c_k = i(I')$ by Lemma 2.4. (Lemma 4.2 does not apply directly: c_0 is not a reflection.) $\square$

4.4 The E_{10} family

The first indefinite rank E_{10} lies outside the sequence E_{4r+1} . Embed the affine subdiagram on $\{0, \dots, 7, 9\}$ and take $\hat{\delta} = (2, 4, 6, 5, 4, 3, 2, 1, 0, 3)$, $\hat{\gamma} = (1, 2, 3, 2, 2, 1, 1, 0, 0, 1)$ (so $\hat{\gamma}$ differs from γ' of Section 4.3), $\hat{\beta}_0 = (3, 7, 10, 9, 7, 6, 4, 3, 1, 6)$, $\hat{I} = \{1, 3, 5, 7, 9\}$, $\hat{T} = r_{\hat{\gamma}} r_{\hat{\gamma} + \hat{\delta}}$, and for $k \geq 0$

$$\hat{\beta}_k = \hat{\beta}_0 - k\hat{\gamma} + (k^2 + 3k)\hat{\delta}, \quad \hat{b}_k = r_{\hat{\beta}_k}. \quad (6)$$

Proposition 4.8. *The reflections $\hat{b}_k$, $k \geq 0$, are pairwise distinct full-support terminal elements of $W(E_{10})$ that are not fully commutative, with $L(\hat{b}_k) = R(\hat{b}_k) = \hat{I}$. For every fully commutative x and every $k \geq 0$, $\mu(x, \hat{b}_k) = 0$.*

Proof. The explicit lowering sequence in Appendix A.3 reduces $\hat{\beta}_0$ to α_9 ; $\hat{\gamma}$, $\hat{\gamma} + \hat{\delta}$ are real roots of the embedded affine subsystem; and $(\hat{\beta}_0, \hat{\gamma}) = -3$, $(\hat{\beta}_0, \hat{\delta}) = -1$, so by Lemma 4.4, $\hat{\beta}_k = \hat{T}^k \hat{\beta}_0$ with nonnegative increments $4\hat{\delta} - \hat{\gamma}$, $2\hat{\delta}$, whence $\hat{\beta}_k > 0$ in every coordinate. The pairings

$$((\alpha_j, \hat{\beta}_k))_{j=0}^9 = (-1, 1, -2-k, 1+k, -1-k, 1+k, -1-k, 1+k, -1-k^2-3k, 2+k)$$

are positive exactly on $\hat{I}$ with nonpositive sums over adjacent entries. Hence both descent sets equal $\hat{I}$, and $\hat{b}_k$ is terminal by (2). Since $(\alpha_1, \hat{\beta}_k) = 1$, column 1 changes every row, so $\hat{b}_k$ has full support. The coordinate $(\hat{\beta}_k)_0 = 2k^2 + 5k + 3$ is strictly increasing, so the roots and their reflections are pairwise distinct. $\hat{I}$ is a maximum independent set of odd size 5 (Remark 2.7) and $\hat{\beta}_k$ has ten nonzero coordinates, so by Lemma 4.2, $\mu(x, \hat{b}_k) = 0$ for every fully commutative x , and $\hat{b}_k \notin W_{\text{FC}}$ by Lemma 4.1(iii). $\square$

Under standard parabolic embeddings into larger E_n , the $\tilde{E}_8$ and E_{10} families remain terminal and have the supports of the smaller parabolic subgroups.

Every fixed E_n has only finitely many FC elements [1, §5.1]; hence sufficiently long members of a pairwise distinct family have no FC covers. For the families above, we exclude covers for every parameter by the reflection and translation arguments. Coefficients at non-covers vanish by maximum common descents and parity. These families have full support in the stated ranks. The classification of all terminals and the bound for arbitrary upper endpoints in infinite type E_n remain open.

5 Complements and open problems

5.1 Necessity of the fully commutative hypothesis

The complete E_6 calculation returns $\max \mu(x, w) = 10$ (Appendix A). One pair attaining this maximum is

$$x = 01254012010, \quad w = 123012523412301251234012.$$

Here $\ell(x) = 11$, x is not FC, and w has full support and length 24. The polynomial is

$$P_{x,w} = 1 + 13q + 59q^2 + 121q^3 + 125q^4 + 61q^5 + 10q^6.$$

Exactly 1,386 pairs have $\mu \geq 2$ (400, 556, 310, 108, 4 and 8 pairs with $\mu = 2, 3, 4, 5, 6, 10$), all with non-FC lower endpoint, while all 6,431 nonzero values with fully commutative lower endpoint equal 1, a brute-force confirmation of Theorem 1.1 for E_6 (by parabolic inclusion, $\mu \geq 10$ occurs in E_7 and E_8 as well). Full commutativity of the upper endpoint alone does not suffice either: $x = 5343010$ (not FC) and $w = 0125342312501234$ (FC, $\ell = 16$) have

$$P_{x,w} = 1 + 8q + 22q^2 + 20q^3 + 5q^4, \quad \mu(x, w) = 5.$$

5.2 The bound fails in affine type

In $\tilde{D}_4$, with x the product of the four leaves and c the centre, $w = xcx$ has $P_{x,w} = 1 + 3q + 2q^2$ and $\mu(x, w) = 2$. The upper endpoint is a boolean reflection: commuting the leaves rewrites it as a palindrome whose first half and centre use distinct generators. By Mongelli's formula for boolean elements in tree Coxeter groups [21, Corollary 3], $f_4(q) = 1 + 3q + 2q^2$. The equality $\mu(x, w) = 2$ also follows directly from Corollary 2 of the same source. The value is recomputed in Appendix A. Hence Corollary 1.2 is sharp within simply laced type, and the universal form of Green's question has a negative answer. Vanishing of μ does not mean $P = 1$: in finite D_4 , with x the product of the three leaves, $b = xcx$ has maximum common commuting descents and even gap 4, so $\mu(x, b) = 0$, but $P_{x,b} = 1 + 2q$ [21, Corollary 3] (the same construction with three leaves gives f_3).

5.3 Terminals in affine E_8 beyond the family

The reflection

$$w = r_\beta = 875634523412301282341230128567453, \quad \beta = (1, 2, 3, 3, 2, 2, 1, 1, 2),$$

of length 33 is a full-support terminal with $L(w) = R(w) = \{3, 5, 7, 8\}$, which is not a maximum independent set, so Proposition 2.5 does not apply. It has exactly five eligible fully commutative lower endpoints, all with $\deg P_{x,w} = 13$:

x	8753	87530	87531	875301	875310
gap $\ell(w) - \ell(x)$	29	28	28	27	27
$\mu(x, w)$	0	0	0	1	1

The right extensions ws_0 and ws_1 of length 34 are terminal involutions with $L = R = \{0, 3, 5, 7, 8\}$, respectively $\{1, 3, 5, 7, 8\}$, each a maximum independent set, each with a unique eligible fully commutative lower endpoint (87530, respectively 87531, gap 29, $\deg P = 14$) and $\mu = 1$, so these two pairs satisfy every hypothesis of Proposition 2.5 except parity. All computed values lie in $\{0, 1\}$. The polynomials and the sizes of the lower ideals and quotients are recorded in the certificate (Appendix A); the computation runs the ordinary Kazhdan–Lusztig recursion [3, Theorem 5.1.7], identifying lower endpoints in the same right coset of $W_{R(w)}$: for fixed w , $P_{x,w} = P_{xs,w}$ for $s \in R(w)$ [3, Proposition 5.1.8]. The computation evaluates ordinary Kazhdan–Lusztig polynomials using coset representatives.

5.4 Open problems

1. Is $\mu(x, w) \in \{0, 1\}$ for every fully commutative x and every $w \in W(\tilde{E}_8)$? All values computed here, including those of Section 5.3, lie in $\{0, 1\}$.
2. Let Λ be the translation subgroup of $W(\tilde{E}_8) = W(E_9)$. Classify the terminal elements modulo the relation $b \sim c$ if $c = tbt^{-1}$ for some $t \in \Lambda$. For indefinite E_n , fix the embedded affine subgroup on $\{0, \dots, 7, n-1\}$ and ask the same question using its translation subgroup. Here the relation is restricted to terminal endpoints; translation conjugation need not preserve terminality in general.
3. Determine μ for the translated length-33 reflection family $r_{\beta+k\delta'}$, $k \geq 1$, of Section 5.3.
4. Determine the bound in F_4 , H_3 and H_4 , the finite irreducible types not treated by the results cited here.
5. Give an effective description of all noncommuting terminals of E_n for $n \geq 9$. The affine family and its standard parabolic embeddings show that no finite list exists.

Use of AI tools

Large language models from OpenAI and Anthropic assisted with literature searches, constructions, proof drafts, implementation, and internal checks. The author directed the work and takes responsibility for the manuscript. The reproducible computations and their dependencies are described in Appendix A; AI-generated review notes are not peer review. The development history is recorded in `PROVENANCE.md` in the repository.

A Computational certificates

A.1 Proof inputs and additional checks

The computational inputs to the finite proof are the exhaustive terminal classification, the word and descent data of Table 1, and the D_6 value

$$P_{x_6, w_6} = 1 + 6q + 11q^2 + 6q^3 + q^4 + q^5.$$

The classification algorithm and its completeness argument are in Section 3.2. Full E_6 and E_7 enumerations in two representations return the same terminal lists as the recursive classification. In E_8 , a separate D_7 -parabolic enumeration and the recursive program along four parabolic chains return the same list. The

D_6 value is checked through all 24,245 dependency records of a saved recurrence certificate and recomputed by separate programs, including directly in the ambient E_7 and E_8 groups.

For the uniform and E_{10} families, the parameter arguments are the symbolic proofs of Section 4. Their finite real-root witnesses are printed in Appendix A.3. For the affine family, we also use the matrix and inversion data of c_0 and the five reduced braid words for its simple descents. Using the translation factorization, we exclude FC covers for every parameter. The complete FC catalogue and the exclusion of all 21 base covers are supplementary checks; the catalogue criterion is proved below. Table 2 identifies the principal programs; the supplement README lists all commands and certificates.

Proof input	Principal commands
Finite terminal lists	<code>terminals-recursive</code> , <code>e8-d7</code> , <code>enumerate-bad</code> , <code>matrix-search</code>
Printed finite words and descents	<code>finite-descents</code> , <code>verify-outputs</code> , <code>verify-e8</code>
D_6 coefficient	<code>d6-certificate</code> , <code>d8-germ-kl</code> , <code>d6-ambient</code>
Uniform root identities	<code>uniform-verify</code>
Affine matrix, inversions and braid witnesses	<code>affine-proof</code>
E_{10} root identities	<code>e10-all-k</code>

Table 2: Principal proof computations. Each name is a command under `supplement/cmd/`. Additional programs check alternative implementations and the supplementary results.

The further computations include the affine FC catalogue and all base covers, checked by `fc-catalogue` and `affine-fc-covers`. They also concern the structure in Remarks 3.4–3.5, the results of Section 5, and checks beyond what the proofs require. These include the finite FC counts (equal to those in [23, Appendix, Table 1] and [1, §5.1]), FC maxima in E_{10} – E_{13} , and finite-rank checks of the uniform identities. The D_8 calculation provides a further check against Gern’s thesis:

$$P_{x_8^D, w_8^D} = P_{e, w_8^D} = 1 + 12q + 59q^2 + 154q^3 + 233q^4 + 221q^5 \\ + 147q^6 + 70q^7 + 20q^8 + 2q^9.$$

Here $x_8^D = s_1 s_2 s_4 s_6 s_8$ and $w_8^D = (1, -8, 3, -6, 5, -4, 7, -2)$ in Gern’s signed-permutation convention. This type- D_8 element is distinct from the exceptional terminal w_8^E of Table 1.

A.2 Why the fully commutative catalogue is exhaustive

The catalogue is constructed by increasing length, starting with the identity. At each length, the program retains exactly the elements satisfying the following criterion.

Lemma A.1 (Catalogue criterion). *In a simply laced Coxeter system, a nonidentity element w is fully commutative if and only if $R(w)$ is a commuting set and ws is fully commutative for every $s \in R(w)$.*

Proof. If w is FC, every reduced prefix is FC and $R(w)$ is commuting, so both conditions follow. Conversely, suppose w is not FC and choose a reduced word containing a braid sts , $m(s, t) = 3$. If this occurrence is followed by another letter, deleting the last letter gives a reduced word with the same braid for some wu , $u \in R(w)$; thus wu is not FC. If the occurrence is at the end, the braid relation replaces the suffix sts by tst , putting both s and t in $R(w)$. In either case one of the two conditions fails. $\square$

For every retained element v of length d , the program forms vs for each right ascent s . A candidate w of length $d + 1$ is retained exactly when its right descents commute and all wt , $t \in R(w)$, are in the completed length- d layer. The program detects duplicate elements by comparing root matrices. Induction using Lemma A.1 proves that each layer contains exactly the FC elements of that length. If a layer has no accepted extensions, no longer FC element exists, since every reduced prefix of an FC element is FC. Thus termination is checked by closure, not imposed by a length cutoff.

In E_9 this procedure returns 44,199 elements, with maximum length 44. A separate verifier checks the catalogue and closure, and tests the elements obtained by single-letter deletions from the reduced word for c_0 .

Exactly 21 distinct deletions have length 26; these are all Bruhat covers by strong exchange, and none belongs to the FC catalogue. The absence of FC covers also follows from the braid-witness proof of Theorem 4.7. The maximum and count equal those in [1, §5.2].

A.3 Explicit real-root witnesses

The finite witnesses used in Section 4 can be checked directly from the simple-reflection rule. If $v = (v_j)$, then s_i changes only coordinate i , replacing it by

$$(s_i v)_i = -v_i + \sum_{j \sim i} v_j.$$

A sequence $i_1 \cdots i_m$ below means *apply s_{i_1} first*, then s_{i_2} , and so on; its action is $s_{i_m} \cdots s_{i_1}$. Thus these sequences list reflections in order of application; Table 1 lists them in product order. Parenthesized strings are concatenated.

For the affine subdiagram on $\{0, \dots, 7, p\}$, with branch p at node 2, set

$$\begin{aligned}\sigma &= 210432165432, \\ \tau &= (210432165432)(p210321432543654765)(p210321432).\end{aligned}$$

Applied to γ and $\gamma + \delta$, respectively, these sequences end at α_p . Take $p = 4r$ for the uniform construction and $p = 9$ for the hatted roots of E_{10} . All intermediate vectors are positive, and each step decreases height. For example, the first step on $\gamma = (1, 2, 3, 2, 2, 1, 1, 0, 1)$ in coordinates $0, \dots, 7, p$ changes coordinate 2 from 3 to $-3 + 2 + 2 + 1 = 2$.

For $\hat{\beta}_0$ in E_{10} , the sequence

$$(135476592103214325678921032)(14325436547659210321432)$$

ends at α_9 , again through positive vectors of decreasing height. By these identities, the vectors are real roots. The verification programs regenerate and check the same witnesses.

A.4 Reproduction and implementation independence

All arithmetic in the supplement is exact. Separate implementations provide checks with different representations. The flat and recursive terminal engines share the library `internal/parabolic`. The commands `affine-proof` and `affine-fc-covers` share `internal/affine`. The saved D_6 recurrence checker shares group arithmetic with its generator, so it is a consistency check. The `d8-germ-kl` and `d6-ambient` recomputations use separate code. The supplement README records the remaining dependencies.

The repository <https://github.com/tygern/kl> contains the source, inputs, certificates and archive builder. From its root, run

```
go run -C supplement ./cmd/package
```

to build an archive containing the current manuscript snapshot and a SHA-256 manifest. The archive is `supplement/dist/exceptional-leading-proof.zip`. Extract it, enter the directory `exceptional-leading-proof`, and run

```
go run ./cmd/proofs -full
```

The runner rebuilds the programs, recomputes in an isolated directory, compares certificates with the supplied snapshots, and checks a stable summary. Go 1.22 or later suffices; no external packages are required. For the finite theorem alone, run `go run ./cmd/proofs -finite`: this regenerates the terminal classifications, checks the printed words and maximum descents, and verifies and recomputes the residual D_6 coefficient. Its summary contains only finite inputs. The flags `-finite` and `-full` are mutually exclusive. The default mode recomputes the classifications, compares saved full-group E_6/E_7 snapshots, and omits heavier cross-checks. Release [v0.6.1](#) contains the revised prose, diagrams, study companion and audits; v0.6.0 contains the preceding proof revision. The manifests record source hashes for integrity checks.

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